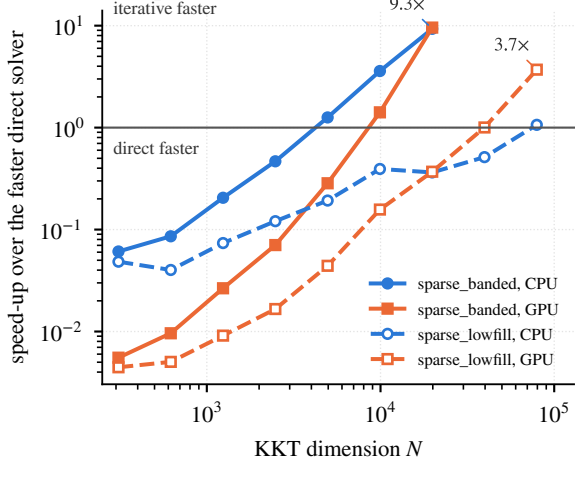
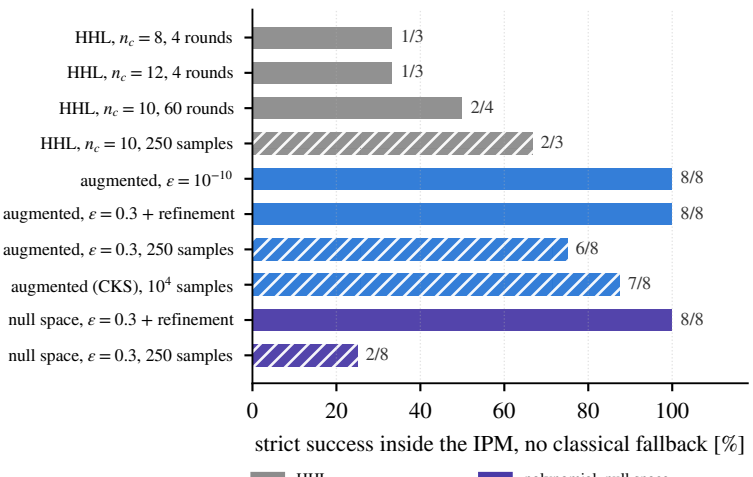

(a) equilibrated MINRES vs. factorisation
iterative faster
direct faster
9.3×
3.7×
speed-up over the faster direct solver
KKT dimension N
sparse_banded, CPU
sparse_banded, GPU
sparse_lowfill, CPU
sparse_lowfill, GPU


(b) simulated quantum KKT solvers
HHL, $n_c = 8$, 4 rounds
HHL, $n_c = 12$, 4 rounds
HHL, $n_c = 10$, 60 rounds
HHL, $n_c = 10$, 250 samples
augmented, $\varepsilon = 10^{-10}$
augmented, $\varepsilon = 0.3$ + refinement
augmented, $\varepsilon = 0.3$, 250 samples
augmented (CKS), 10^4 samples
null space, $\varepsilon = 0.3$ + refinement
null space, $\varepsilon = 0.3$, 250 samples
1/3
1/3
2/4
2/3
8/8
8/8
6/8
7/8
8/8
2/8
strict success inside the IPM, no classical fallback [%]
HHL
polynomial, augmented
polynomial, null space
finite-shot readout